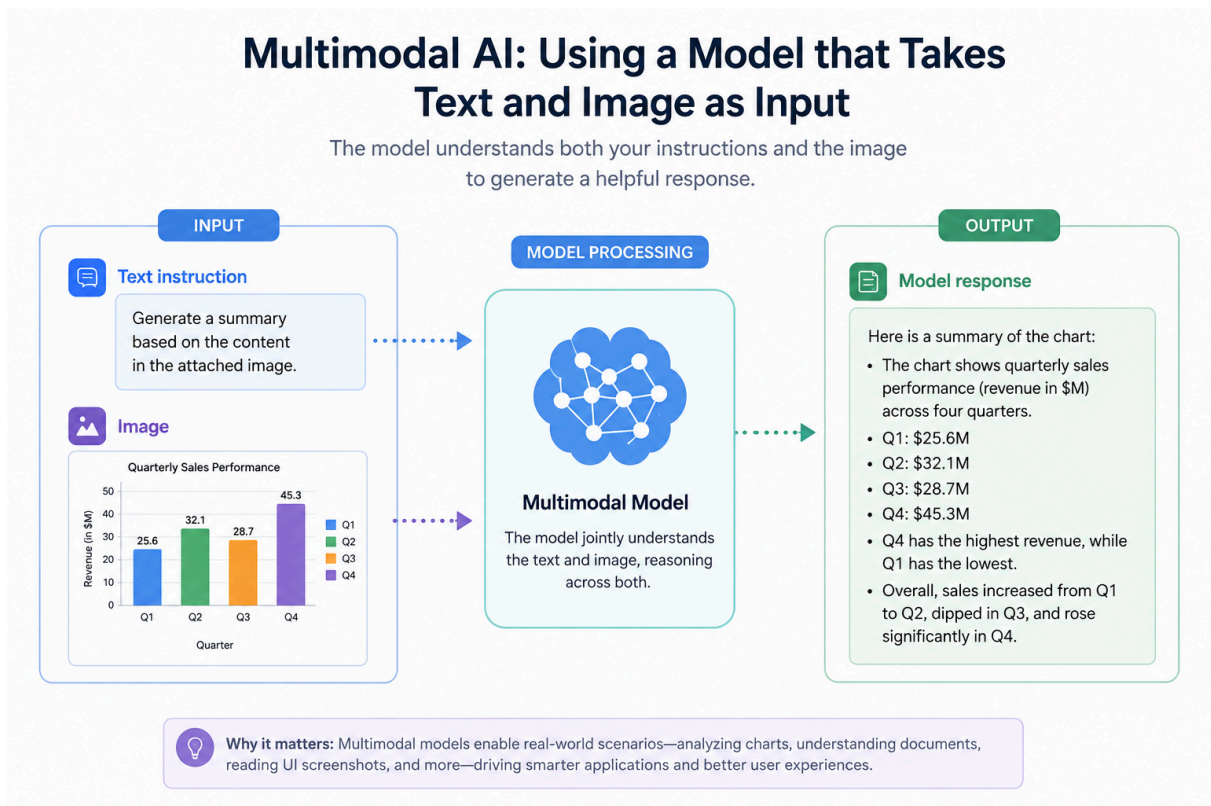




Implement computer vision solutions

Lab - Sending Images as Context - Vision Input on Chat Models





In this lab, we are looking at how to send an image as context to a chat model, so the model can reason over both the user's text instruction and the visual information in the image.

Here, the input has two parts. The first part is the text instruction, where we tell the model what we want it to do. For example, we might ask the model to generate a summary based on the content in the attached image. The second part is the image itself, such as a chart, screenshot, document, or UI screen. Instead of describing everything manually in text, we allow the model to inspect the image directly.

The key idea is that the model is multimodal. This means it can understand more than one type of input. In this case, it processes the written instruction and the image together. So if the image contains a chart, the model can identify the values, compare the data points, understand the overall trend, and then generate a useful response based on the question we asked.

In the example shown, the user provides a quarterly sales chart and asks for a summary. The model reads the chart, extracts the sales values for each quarter, identifies that Q4 has the highest revenue, and explains the trend across the year. This is much more powerful than a simple text-only interaction, because the model is not just answering from text — it is interpreting visual content.

This matters in real business scenarios because many useful inputs are visual. Teams often work with charts, invoices, screenshots, support tickets, dashboards, architecture diagrams, and reports. By sending images as context, we can build applications where users upload visual information and receive clear, helpful explanations from the model.

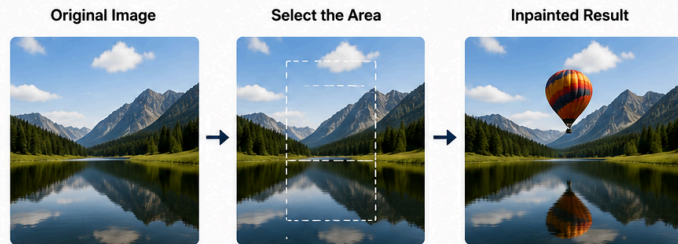
Image editing with generative models

Inpainting

Inpainting means editing a selected part of an image while keeping the rest of the image mostly unchanged.



You give the model an existing image and say, **“Change only this part.”** The model fills in that region in a way that visually fits the surrounding image.



Real-world Examples

Add an object

Add a coffee cup beside the laptop.



Replace the background

Change to a modern office setting.



Remove a distracting object

Remove the water bottle from the table.



Why it matters for businesses

Businesses usually work with existing assets—brand images, product shots, diagrams, UI screenshots, and marketing visuals. Inpainting lets them reuse those assets instead of starting from zero.



Brand Images



Product Shots



UI Screenshots



Marketing Visuals



In this chapter, we are looking at how generative models can be used not only to create new images, but also to edit existing images in a controlled way.

The first concept is **inpainting**. Inpainting means changing only a selected part of an image while keeping the rest of the image mostly unchanged. Instead of regenerating the entire picture from scratch, we give the model an existing image and tell it what part should be modified. The model then fills that selected area in a way that visually matches the surrounding image.

For example, we might start with a landscape image and ask the model to add a hot air balloon in the sky. The important point is that the mountains, lake, reflection, lighting, and overall style should remain consistent. The model is not just pasting an object randomly into the image. It tries to blend the new object naturally into the scene.

This same idea can be used in many practical scenarios. We can add an object, such as placing a coffee cup beside a laptop. We can replace a background, such as changing a plain desk setup into a modern office environment. We can also remove distracting objects, such as a water bottle on a table, while preserving the rest of the image.

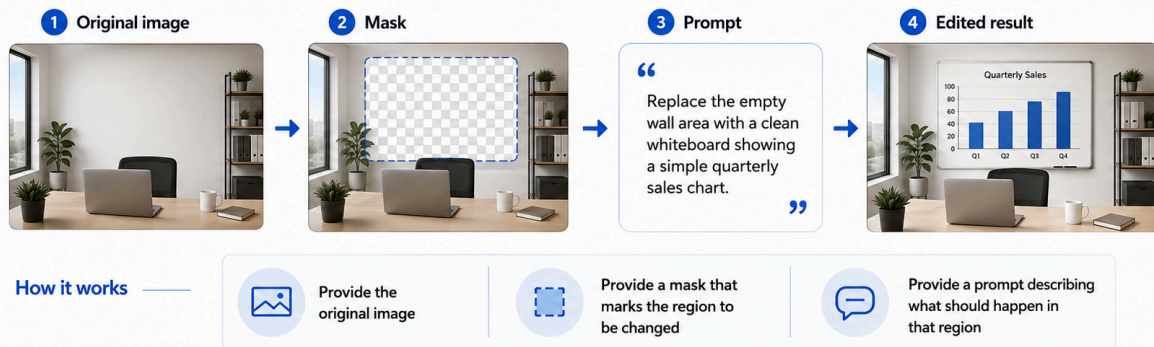
This matters because businesses often already have existing visual assets. They may have product photos, brand images, UI screenshots, marketing visuals, or training diagrams. With image editing, they do not always need to recreate everything from the beginning. They can take an existing image and make targeted changes, which saves time and keeps the visual style more consistent.

Mask-based edits

Mask-based editing is a more controlled form of inpainting. A mask tells the model exactly which pixels are editable, so the rest of the image can stay unchanged.



In Azure OpenAI image editing, the mask must be a **PNG** with the **same dimensions** as the input image. Fully transparent pixels indicate the area the model should edit.



Why masks matter

When precision matters, a mask helps localize the edit. Without a mask, the model may interpret the request more broadly and change more of the image than expected.



Precision
Edit exactly what you intend.



Localized edits
Focus changes only where you specify.



Fewer unintended changes
Keep the rest of the image intact.



The second concept is **mask-based editing**. This is a more controlled form of image editing. A mask tells the model exactly which pixels are editable and which pixels should remain unchanged. This is useful when we want the model to focus only on a specific region of the image.

In the example shown, the original image contains an empty wall behind a desk. A mask is placed over that wall area to indicate that this is the region we want to change. Then we provide a prompt, such as asking the model to replace the empty wall with a clean whiteboard showing a simple quarterly sales chart. The model then edits only that masked region and produces the final image.

The key benefit of using a mask is precision. Without a mask, the model may interpret the instruction more broadly and make changes to parts of the image that we did not intend to modify. With a mask, we can localize the edit, keep the rest of the image intact, and reduce unwanted changes.

Lab - Building an Azure AI Content Safety resource

Azure AI Content Safety - <https://learn.microsoft.com/en-us/azure/ai-services/content-safety/overview>